lit-acquisition: multi-source, multilingual biomedical literature acquisition with citation-graph traversal

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Summary

Comprehensive literature review is essential for clinical variant interpretation, yet most retrieval tools query one database and are English-centric. **lit-acquisition** unifies **19 literature channels** — 15 federated search backends plus PubMed, Semantic Scholar, ClinicalTrials.gov and Zenodo services — behind one interface, and adds LLM-based cross-lingual query translation and citation-graph traversal. On an exploratory reference set of **43 genetic variants (822 papers)** that is heuristically annotated, not expert-verified, and whose candidate pool overlaps the evaluated system, the end-to-end pipeline raises PubMed-only set recall from **0.269 to 0.467** (+0.198; bootstrap 95% CI [+0.123, +0.269]; Holm-adjusted $p = 0.0002$), and recovers **21 of 24** annotated German/French papers where every English-only baseline finds at most one. The toolkit ships as a lightweight API-only edition and an LLM-enhanced agent edition (v0.5.0, MIT).
Availability: `pip install lit-acquisition` · `https://github.com/lanshi17/lit-acquisition` · MIT licence.
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1 Introduction

Variant classification under the ACMG/AMP framework requires the assembler to marshal published evidence: case reports, functional assays and segregation data that together support criteria such as PS3 or PVS1 [1]. That evidence is scattered across dozens of databases, publishers, preprint servers and languages, and it is growing faster than any curator can read.

Existing tooling leaves three gaps. *Single-source* interfaces (PubMed, Europe PMC) see only their own index, while a variant's literature is split across Crossref-registered journals, PMC deposits and non-indexed venues [3, 4, 5]. *Language*: clinically decisive case reports appear in German, French, Japanese or Russian venues that English-only queries cannot reach. *Citation structure*: keyword search misses papers whose titles omit the variant, although they are one hop away. Commercial curation platforms cover parts of this, but they are closed and expensive; no open toolkit combines all three behind one scriptable interface.

We present **lit-acquisition**, an open-source Python toolkit that closes those three gaps in one pipeline, and we report an exploratory coverage evaluation against a PubMed-centred reference set with an explicit account of what that reference set can and cannot support.

2 Implementation

Pipeline

Acquisition runs in three phases (Figure 1A). *Search* fans out across 19 channels — 15 pure-Python `httpx` backends (Crossref, OpenAlex [6], Europe PMC, PMC, DOAJ, J-STAGE, CiNii, SciELO, BASE, CORE, OpenAIRE, arXiv, bioRxiv, medRxiv, Unpaywall) plus PubMed, Semantic Scholar, ClinicalTrials.gov and Zenodo clients — with a shared connection pool, per-provider deadlines, health-aware ordering and a circuit breaker; results are deduplicated on DOI, PMID and normalised title. *Download* resolves open access only (DOI→Unpaywall, PMCID→Europe PMC render, then direct URL) with SSRF validation on every redirect hop and streamed size caps. *Gate / rank* applies token-overlap ranking (CJK-bigram aware) and, in the agent edition, a cross-encoder rerank [11] plus an LLM relevance gate.

Multilingual routing and citation traversal

The agent edition translates the query into six languages (en/zh/ja/de/fr/ru) with domain-aware prompts that preserve gene symbols and HGVS nomenclature, then routes each language to the channels that index it; the recorded run used Qwen2.5-72B-Instruct at temperature 0.2, with the prompt hash stored in the run manifest [12]. From a seed DOI the toolkit additionally walks Semantic Scholar citation and reference edges breadth-first (depth and fan-out configurable; depth 1, 20 papers here), deduplicating against keyword results [7].

Packaging

Python ≥3.12, on `httpx`, `pydantic`, `loguru` and `pymupdf`. `pip install lit-acquisition` installs the lite edition; `lit-acquisition[agent]` adds the OpenAI-compatible client for translation, gating and reranking, which the lite edition replaces with coded warnings (`CONFIG_MISSING`, `PROVIDER_HTTP_429`, `TIMEOUT`, ...) and a purely heuristic pipeline.

Table 1: Relative coverage over 43 variants (macro means). M4n is the nested ablation: frozen M3 ranking plus citation-graph items.

	SetRecall	Recall@20	nDCG@20	Relevant
M1 PubMed-only	0.269	0.171	0.205	213
M2 multi-source	0.472	0.153	0.164	373
M3 + translation	0.465	0.276	0.301	392
M4n + citation	**0.467**	**0.276**	**0.301**	**394**

3 Results

Reference set and protocol

The reference set holds 43 clinically actionable variants (BRCA1, TP53, CFTR, HFE, ...) and 822 papers, built by an automated heuristic protocol scoring gene, variant token, disease keyword and publication type. It is *not* expert-verified and its candidate pool was drawn from channels that overlap the evaluated system, so every number below is a *relative* coverage comparison, not a clinical retrieval-quality estimate. An offline language backfill raises the German bucket from 13 to 18 papers and the French from 11 to 20 (Suppl.). We compare four configurations at a uniform 100-candidate budget with a strictly nested ablation; paired differences use two-sided Wilcoxon signed-rank with Holm–Bonferroni correction over a pre-declared family of eight tests (10,000 bootstrap resamples, seed 42).

Federation is the main effect

The pipeline significantly improves coverage over PubMed-only: SetRecall +0.198 (95% CI $[+0.123, +0.269]$; $p_{\text{Holm}} = 0.0002$; matched-pairs rank-biserial $r = 0.74$), and shallow ranking improves as well (Recall@20 +0.105, $p_{\text{Holm}} = 0.016$; nDCG@50 +0.162, $p_{\text{Holm}} = 0.0004$). Decomposing the gain shows that *federation* carries it (+0.203 over M1, $p_{\text{Holm}} = 0.0001$): translation and citation are set-level neutral ($p_{\text{Holm}} = 1.0$ in both pairwise contrasts), and citation traversal costs list precision (−0.011 Precision@Retrieved, $p_{\text{Holm}} = 1.9 \times 10^{-6}$).

Multilingual recovery

Translation does not raise aggregate set recall, but it changes *which* papers are reachable: M3/M4n recover 21 of the 24 annotated German and French reference papers (11/13 German, 10/11 French; 16/18 and 18/20 after backfill), whereas every English-only baseline — including multilingual dense retrieval — recovers at most one. This is the clearest practical argument for language-aware routing.

Citation traversal and single-channel baselines

Across all 43 variants citation traversal contributes two relevant papers that keyword search did not rank, one appearing in no keyword list at all (V002 BRCA1 c.5266dupC; V035 HFE p.C282Y) — a bounded complementary channel, not a headline gain. Against six 2025–2026 single-channel techniques (dense retrieval [9], cross-encoder reranking [10], hybrid BM25 [8]+dense fusion, LLM query expansion, agentic two-round search, multilingual dense) the pipeline is significantly better than five; the strongest, hybrid BM25+dense (0.421), stays below M4n (0.467) but not significantly so (raw $p = 0.35$, Holm-adjusted $p = 1.00$). Four of the six only reorder the same PubMed pool (mean 80.7 candidates) and are identical to M1 on set recall by construction; re-applying every ranker to the frozen pools with a corrected cross-encoder leaves the picture unchanged (Suppl.).

Cost

In the 43-variant session the composed full pipeline took a median of 28.65 s per variant (p90 44.61 s) at a 100-candidate budget, including provider back-off. A separate four-variant download measurement landed 18 PDFs from 187 candidates: 3.99 PDFs/min end-to-end, 9.6% candidate-to-PDF, 15.06 s per PDF, peak RSS 85.7 MB. Downloading requires `LIT_UNPAYWALL_EMAIL`; without it every DOI→OA route is skipped.

4 Discussion

The pipeline's coverage gain over PubMed-only is statistically robust here. Two limitations matter. The reference set is heuristically annotated with an overlapping candidate pool, so the deltas measure relative reach, not clinical completeness: against an independent ClinVar-citation truth the gain disappears, because that truth is a decade older than what the providers return (Suppl.). Expert re-annotation is in progress. And the cost is real (29 s per variant, 4 PDFs/min): adequate for curation-scale batches, not for interactive search. Future work targets semantic search, adaptive traversal and integration with variant databases such as ClinVar [2].

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[To be completed.]

Conflict of interest

None declared.

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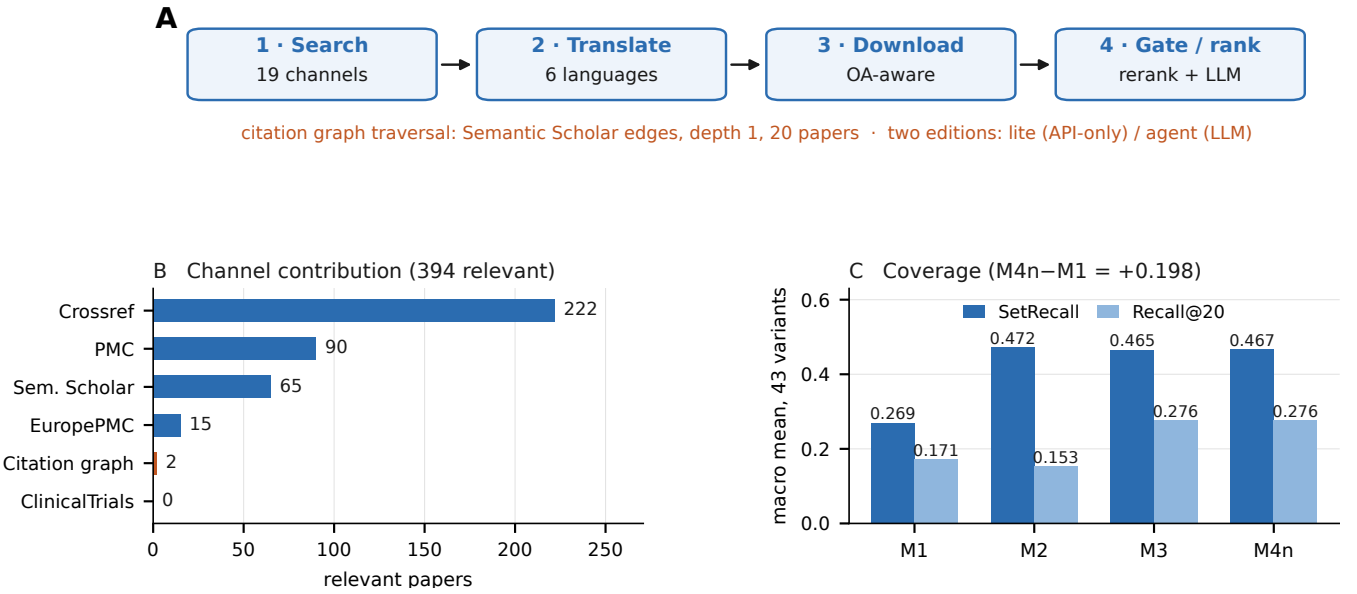


Figure 1: lit-acquisition at a glance. (A) The three-phase pipeline across 19 channels and the two editions. **(B)** Relevant reference papers per channel in the nested M4 run (first-seen attribution after deduplication). **(C)** Macro-mean SetRecall and Recall@20 over 43 variants for M1 PubMed-only, M2 multi-source, M3 +translation and M4n +citation (nested); M4n−M1 = +0.198 (95% CI $[+0.123, +0.269]$, $p_{\text{Holm}} = 0.0002$).

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